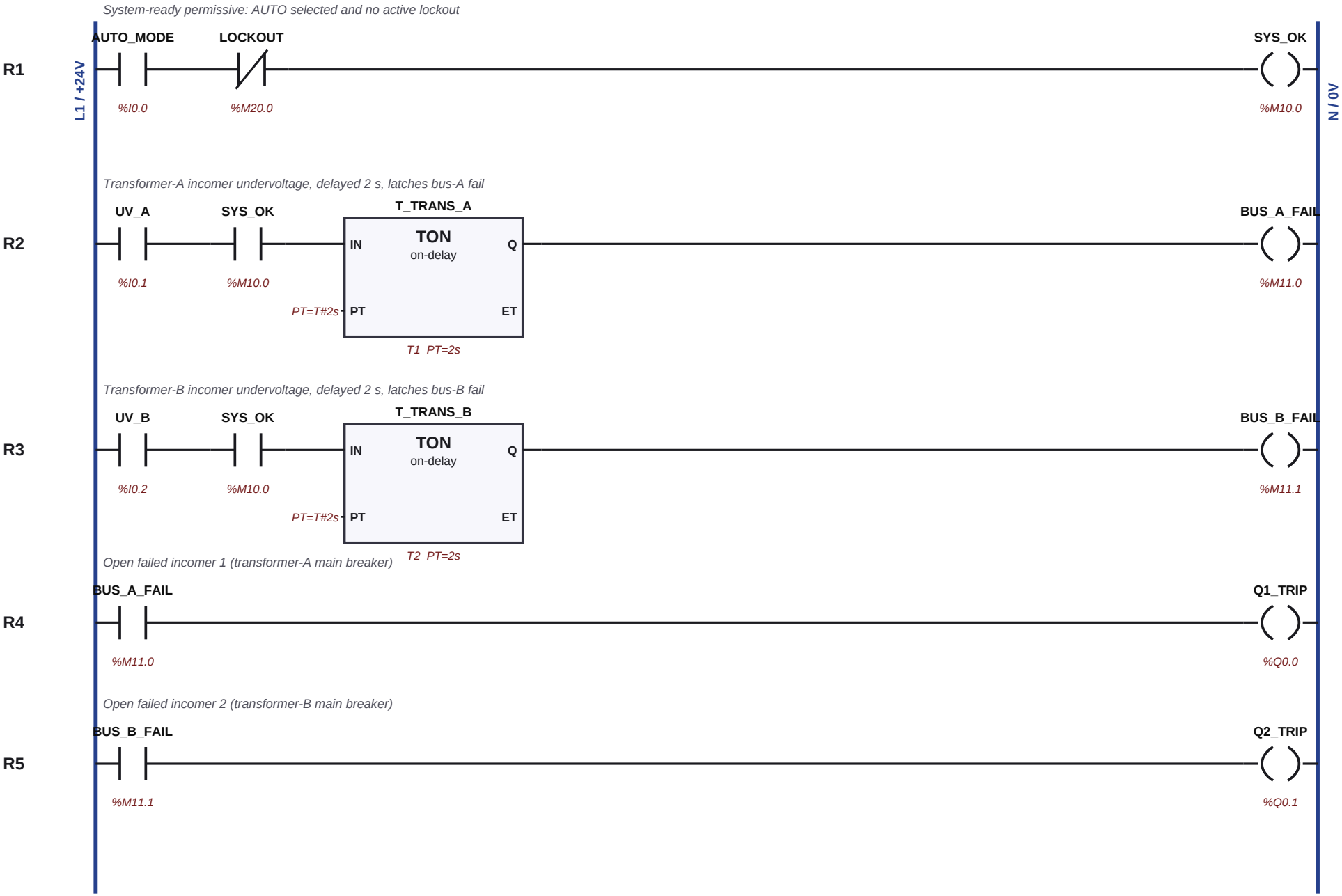
ATS — 2 Transformers + Bus Coupler (Main-Tie-Main)

Automatic Transfer Switch – dual-incomer LV switchboard, open-transition bus-tie

DWG No.	REV	DATE	SHEET
ATS-LAD-001	A	2026-10-21	1 of 2



LEGEND / SYMBOLS

NO contact] [— NC contact]/[— Output coil ( ) — Set/latch (S) — Reset (R)

Addresses: %I = digital input (DI) %Q = digital output (DO) %M = internal memory bit T# = TON on-delay timer Tag shown above element, PLC address below.

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DWG No.	REV	DATE	SHEET
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L1 / +24V

N / 0V

R6

Close bus-tie (Q3) on either bus fail, only if an incomer is already open

BUS_A_FAIL %M11.0 — Q1_CL %I1.0 — BUS_B_FAIL %M11.1 — Q2_CL %I1.1 → Q3_CLOSE %Q0.2

NOTE (2-of-3 interlock): the bus-tie may close only when at least one incomer breaker is open — never with all three breakers (Q1, Q2, Q3) closed, which would parallel the transformers.

R7

Permit to reclose incomer 1 after transformer-A recovers

RETRANS_PERMIT %M12.0 — Q3_CL %I1.2 — RECOV_A %I0.3 → Q1_CLOSE %Q0.3

R8

Permit to reclose incomer 2 after transformer-B recovers

RETRANS_PERMIT %M12.0 — Q3_CL %I1.2 — RECOV_B %I0.4 → Q2_CLOSE %Q0.4

R9

Retransfer timing: confirm source A healthy for 5 s, then open-transition trip of tie

UV_A %I0.1 — RETRANS_PERMIT %M12.0 — T_RETRANS TON on-delay (IN, Q, PT=T#5s, ET) T3 PT=5s — T_RETRANS.Q T3.Q — RECOV_A %I0.3 → Q3_TRIP %Q0.5

Open-transition (break-before-make): the tie (Q3) opens before the recovered incomer recloses.

R10

Lockout latch — FAULT sets, RESET clears (seal-in broken by reset)

FAULT %I2.0 → LOCKOUT (S) %M20.0

RESET %I2.1 → LOCKOUT (R) %M20.0